

# BRAD CAO

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## EDUCATION

### Columbia University

B.A. in Mathematics & Computer Science

New York, NY

Expected May 2029

**Relevant Coursework:** Data Structures in Java, Probability Theory, Natural Language Processing, Honors Mathematics A & B

## EXPERIENCE

### Research Fellow

Sep. 2026 – Present

SPAR (Supervised Program for Alignment Research)

Remote

- Building datasets that evaluate human+AI judge systems for scalable oversight with Rishub Jain and Joshua Jacob (ex-Google DeepMind): a 100+ dataset leaderboard spanning deception, jailbreak robustness, cultural bias, and computer-control mistakes
- Evaluating existing judge datasets for onboarding and creating new ones for taxonomy gaps, with sandwiching for ground truth; measuring judge robustness to reward hacking under adversarial RL optimization

### Data Science Intern

Apr. 2026 – Present

Mercor

San Francisco, CA

- Engineered evaluation frameworks that formalize subject-matter expertise into RLHF-compatible annotation schemas for LLM fine-tuning pipelines
- Designed rubric-based scoring taxonomies and QA workflows to raise inter-annotator agreement; audited quality and flagged edge cases to refine labeling guidelines

### Research Assistant

Apr. 2026 – Present

Duke University

Durham, NC

- Trained and developed LLMs to identify and quantify inter-network disparagement in televised news media, building NLP pipelines to ingest, segment, and label large transcript volumes
- Benchmarked prompt-engineered and fine-tuned classifiers against human-coded samples with Dr. Philip Napoli, translating media-studies frameworks into quantifiable schemas

### Research Intern

Jun. 2025 – Aug. 2025

University of Notre Dame

South Bend, IN

- Awarded a full-ride scholarship for original quantitative research on the pharmaceutical industry's impact on public health policy
- Applied R-based regression and visualization to link industry practices to poverty outcomes; presented at a symposium

### Research Intern

Jun. 2023 – Sep. 2025

University of Maryland, Baltimore

Baltimore, MD

- First-authored a supervised ML model with Prof. Allison Dunn predicting ADHD-related patient safety risks from real-world data
- First-authored a study modeling tacrolimus dose requirements and transplant outcomes in lung transplant recipients on systemic azole antifungals
- Presented at the ASCPT Annual Meetings (2024, 2025); earned the Top Poster Award (May 2025)

## PROJECTS

### Threshold | Python, NumPy, SciPy, PyTorch | [github.com/bradca0/threshold](https://github.com/bradca0/threshold)

- Reproduced double descent on MNIST with a random-feature model (nested random ReLU features, min-norm least squares via SVD/Gram decompositions in float64;  $n = 2,000$ , 5 seeds): test error peaks at  $p/n = 1$  (88.8% vs. 24.3% at  $p = 5n$ ; 20% label noise)
- Explained the peak:  $\|\beta\|_2$  and the smallest singular value of  $\Phi$  are extreme at  $p = n$  ( $\|\beta\|_2$ : 811 vs. 2.6 at  $p = 5n$ ); a bias-variance split over 10 draws puts 90% of peak error in variance; tuned ridge cuts the peak from 88.8% to 12.3%
- Showed more data can hurt (fixed  $p = 2,000$ : test error 34.8%  $\rightarrow$  89.1% as  $n$  grows 250  $\rightarrow$  2,000) and replicated peak and mechanism on Fashion-MNIST; a width-swept MLP showed no peak, reported as a negative result; 85 tests, 94% coverage, `mypy --strict`

### MyCureHope | Django, PostgreSQL, JavaScript, TailwindCSS | [mycurehope.org](https://mycurehope.org)

- Co-developed an open-source patient database of treatments, doctors, and hospitals serving 33,000+ users across 78 countries; own search build and technical development
- Built and deployed the full-stack platform (Django, PostgreSQL, JavaScript, TailwindCSS) on Nginx/Gunicorn/Ubuntu

## PUBLICATIONS

Cao, B., Lee, S., Naga, A., Gobburu, J., Dunn, A. "Leveraging Supervised Machine Learning to Identify Predictors of Serious Clinical Outcome with Stimulant Treatment from Real-World Data." *ASCPT Annual Meeting*, 2025.

Cao, B., Gupta, P., Dunn, A. "Predictive Modeling of Tacrolimus Dose Requirements and Transplantation Outcomes in Lung Transplant Recipients on Systemic Azole Antifungals." *ASCPT Annual Meeting*, 2024.

## TECHNICAL SKILLS & INTERESTS

**Languages:** Python, JavaScript, Java, SQL (PostgreSQL), R, HTML/CSS

**Frameworks & Tools:** PyTorch, TransformerLens, NumPy, pandas, scikit-learn, Django, TailwindCSS, Nginx, Gunicorn, Ubuntu/Linux

**ML & Data:** Mechanistic interpretability, LLM fine-tuning, RLHF annotation, supervised ML, data analysis

**Interests:** Fencing (USA Fencing First Team All-American), Competitive Programming (USACO), History Olympiads, Open-Source